

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Final Exam, Spring 2025

Course Code: CSE 4743

Course Title: Computer Security

Time: 2 hours 30 minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes mentioned in additional Columns

Part A

[Answer the questions from the followings]

1. a) A student developer is building an RSA encryption module. To design the key generation part, she recalls the fundamental theorem of arithmetic. Explain how this theorem justifies the use of prime factorization in RSA. CO1 2
- b) In a secure chat app, two users use Diffie-Hellman to establish a shared key. Describe the step-by-step process they follow to derive the shared secret over an insecure channel. CO2 3
- c) An email client encrypts a message using the sender's private key and then the receiver's public key. Use a diagram to explain how this achieves both authentication and secrecy. CO2 5
- OR
- A) A bank application uses symmetric encryption for transactions and public-key encryption for customer registration. Differentiate between these two schemes and explain why public-key is used in registration. CO1 2
- B) In a cryptography class, you are given $p = 17$ and $q = 11$. Demonstrate how you would generate the public key (e, n) and private key (d, n) for an RSA encryption system. CO2 3
- c) A government portal allows citizens to digitally sign forms. Explain how digital signatures ensure integrity and authenticity in such applications with an appropriate diagram. CO2 5
2. A) Explain the design and steps of Secure Hashing Algorithm-3 (SHA-3). CO2 5
- b) Explain how digital certificate is used as a trust mechanism to verify the owner of the certificate. CO2 5

Part B

[Answer the questions from the followings]

3. a) Explain the requirements for a cryptographic hash function. CO3 5
- b) Elaborate how confidentiality and authentication is ensured in S/MIME. CO3 5
4. a) What is the purpose of Kerberos protocol? Elaborate the steps of Kerberos protocol. CO3 5
- b) Differentiate between tunnel and transport mode IPSEC. CO4 5

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring-2025
Course Code: CSE-4871 Course Title: Neural network and Fuzzy System
Total marks: 50 Time: 2 hours 30 minutes.

M CO DL

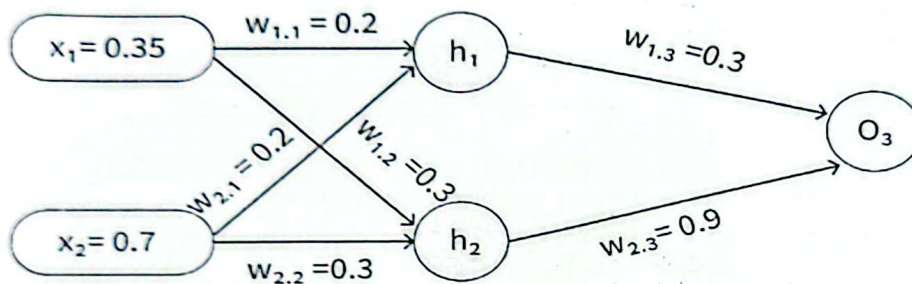
5. a) Propose a cryptographic mechanism (e.g., hashing) to secure blockchain transaction data. Show how it integrates into the blockchain data structure. CO3 4
- b) Design a blockchain-based land registration system that ensures data immutability, confidentiality, and distributed trust. Explain how consensus and encryption mechanisms will be used. CO3 6
- OR**
- ✓) Compare a blockchain-based e-voting system with a centralized voting system. Identify how blockchain solves issues of security, transparency, and scalability. CO3 4
- b) Choose and justify a suitable consensus algorithm (e.g., Proof-of-Work, Proof-of-Stake) for a cryptocurrency system. Evaluate trade-offs in energy efficiency, decentralization, and security. CO3 6

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 Total marks: 50 Time: 2 hours 30 minutes

M CO DL

Group-A

- 1 a) Assume the neurons use the sigmoid activation function for the forward and backward pass. The target output is 0.5 and the learning rate is 1. Apply the Backpropagation algorithm where you need to apply at least two forward passes and one backward pass. Terminate your calculation when you got $Y_5 = 0.61$ in second forward pass (Required steps: forward pass > backward pass > forward pass > Stop). 10 3 A



- 2 a) Explain the concept of Gradient Descent in neural networks? Describe its role in minimizing the loss function during training. 5 2 A
- 2 b) Discuss the importance of regularization in neural networks. Justify the importance of learning rate. How can you adapt to set the learning rate while a Neural Network (NN) is being trained? 5 2 A

OR

2 a)

Observation	True Label (y)	Predicted Probability (p)
1	1	0.9
2	0	0.2
3	1	0.8
4	0	0.4

Calculate the average BCE loss and write your comment whether the model's predictions are good or poor based on the loss values.

2 b)

Example	True Class (One-Hot)	Predicted Probabilities
1	[0, 1, 0]	[0.1, 0.8, 0.1]
2	[1, 0, 0]	[0.7, 0.2, 0.1]
3	[0, 0, 1]	[0.2, 0.3, 0.5]

Calculate the average CCE loss over the 3 examples. Interpret whether the model's predictions are good or poor based on the loss values.

Group-B

- 3 a) Explain the architecture of an LSTM (Long Short-Term Memory) network. Illustrate with a diagram showing how information flows through an LSTM cell. 7 3 A

OR

- 3 b) A Convolutional Neural Network (CNN) is designed for an image classification task. The input image has a shape of (64, 64, 3). Determine the activation shape and number of parameters for each layer in the network. 7 3 A

	Activation shape	Activation Size	No of parameters
Input	(64,64,3)		
CONV1(f=5, s=1)	(60,60,16)		
POOL1	(30,30,16)		
CONV2(f=3, s=1)	(28,28,32)		
POOL2	(14,14,32)		
FC3	(120,1)		
FC4	(60,1)		
Softmax	(5,1)		

Here, CONV = Convolutional Layer (f = filter size, s = stride), POOL = Pooling Layer (2x2 with stride 2), FC = Fully Connected Layer, Softmax = Output Layer for classification into 5 categories.

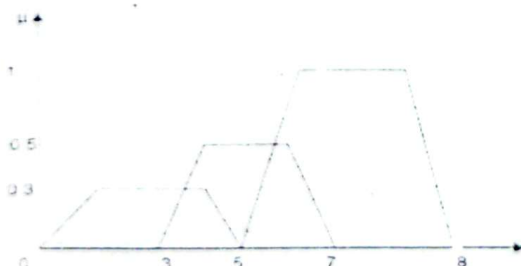
- 3 c) As you train your model you realize that you don't have enough data. Describe three augmentation technique that can be used to overcome the shortage of data. 3 2 U
- 4 a) What is Fuzzy in a Fuzzy Set? How do you obtain a Fuzzy set from a crisp value? 2 2 U
- 4 b) Design a fuzzy logic controller(FLC) for an Automated grading of mango with proper membership function. Consider three input and one output and write detailed rules 5 3 A
- 4 c) What is membership function? State the features of membership function. 3 1 R

OR

Using your intuition assign the membership functions for

a) Population of People b) Employment Strategy c) Usage of Library

- 5 a) Write the name of methods of Defuzzification. Explain Center of Sums(COG) method with example. 3 1 R
- 5 b) Calculate the defuzzified value for the following fuzzy output using weighted average method. 5 3 A



- 5 c) Describe different types of maxima method. 2 1 U

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE
Final Exam, Spring 2025

Course Code: MGT-3601

Time: 2 hours 30 minutes

Course Title: Industrial Management

Full Marks: 50

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

- | | | | | |
|-------|--|-----|----|---|
| 1. a) | What is the purpose of organizational control? Why is it important? | CO3 | An | 5 |
| 1. b) | How would you differentiate between Bureaucratic control and Decentralized control system? | CO2 | E | 5 |
| 2. a) | Marketing manager seek to influence the level, timing and and composition of demand to meet the organization's objectives. For demand management marketer can identify eight possible demand state of the consumers. Identify and discuss those 8 demand states. | CO1 | U | 5 |
| 2. b) | For setting marketing strategy marketer follow different marketing philosophies. Selling concept and Marketing concept are two concepts among them. Describe these two concepts. | CO2 | An | 5 |

Or

- | | | | | |
|-------|--|-----|----|---|
| 2. a) | What should be the considering factors in selecting media for advertising? | CO1 | U | 5 |
| 2. b) | Describe the 4Ps of marketing mix. | CO2 | An | 5 |

Part B

[Answer the questions from the followings]

- | | | | | |
|-------|--|-----|---|----|
| 3. a) | Production means transformation of inputs into output. There are six types of transformation process. Discuss them. | CO1 | U | 5 |
| b) | What do you mean by Plant Layout? What should be the considering factors in designing a manufacturing factory layout? Discuss. | CO3 | E | 5 |
| 4. | Analyze the essential elements of a valid contract. | CO2 | R | 10 |
| 5. | Write short notes on:
Write short notes on: Bench Marking, Human resource control, Social Media advertising, and Management by shura. | CO3 | U | 10 |

OR

Write short notes on:

- | | | | |
|---|-----|---|----|
| Office tanning in Ramadan, Process Layout, Benchmarking, Persuasive Advertising | CO3 | U | 10 |
|---|-----|---|----|

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE Spring-2025, Midterm Exam

Title: Social, Professional & Ethical Issues in Computing

Course Code: CSE-4805, Full Marks: 30 Time: 1 Hours 30 Minutes

Instructions to the Candidate: Answer All Questions

- | | Marks | CLO |
|---|-------|------|
| 1. a) Computers have evolved significantly from large mainframes to modern personal devices, with advancements in hardware and software shaping their development. Early computers relied on punch cards, whereas today's systems use powerful processors and cloud computing. Discuss how have developments in computer hardware and software influenced the evolution of modern computing? | 5 | CLO1 |
| b) How A school is considering integrating AI-powered tools into its teaching methods. These tools would provide personalized learning experiences for students, track progress, and help teachers identify areas where students need additional support. However, some teachers are concerned that AI might replace human interaction and reduce the personal touch in education.

Analyze the potential benefits and challenges of using AI in education. How would you recommend balancing technology with human involvement to enhance the learning experience? | 5 | CLO3 |
| OR | | |
| 1. Discuss the applications and impact of computer on:
i. education,
ii. business | 5+5 | CLO3 |
| 2. a) With the rise of the internet and digital databases, personal information has become easier to collect, store, and share. In the past, privacy was protected by physical barriers, but today online tracking, surveillance, and data breaches pose new challenges.

How has computer technology changed the concept of privacy over time, and what ethical concerns does this raise? | 5 | CLO1 |
| b) A popular video streaming platform collects and analyzes user watch history, search preferences, and interaction data to recommend content and improve user experience. It also shares this data with advertisers to generate revenue.

Evaluate the ethical and business implications of using user data for profit without direct user compensation. Propose a fair approach that balances user privacy and platform sustainability. | 5 | CLO3 |
| 3. a) Briefly discuss the current scenario of e-commerce in Bangladesh. How proper enactment of ICT act can protect consumer and suppliers of e-commerce in Bangladesh? | 5 | CLO4 |
| b) Social media platforms like Facebook, Twitter, and YouTube face the challenging task of moderating content to prevent offensive and hate speech. While these platforms aim to protect users, their moderation practices often raise ethical concerns regarding freedom of expression.

How can these platforms balance preventing harm with respecting freedom of expression? Discuss the consequences of excessive censorship versus insufficient moderation. | 5 | CLO2 |

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Final Examination, Spring 2025

Course Code: CSE-4875 Course Title: Pattern Recognition and Image Processing

Total Marks: 50 Time: 2 Hours 30 Minutes

[Answer all the following Questions. Figures in the right-hand margin indicate full marks]

CO Description

- CO1 Understand basic image processing techniques for solving real problems
- CO2 Apply and demonstrate image processing techniques for solving problems in computer science
- CO3 Evaluate algorithms for higher level image processing.

Group - A

- 1(a) Is it correct to say that data consists of both useful information and redundant parts? Briefly justify your answer with an example. 3 CO1 D2
- 1(b) Finds the connected components from the following binary image using appropriate algorithm. 4 CO2 D3

1	1	0	1	1	1	0	1
1	1	0	1	0	1	0	1
1	1	1	1	0	0	0	1
0	0	0	0	0	0	0	1
1	1	1	1	0	1	0	1
0	0	0	1	0	1	0	1
1	1	0	1	0	0	0	1
1	1	0	1	0	1	1	1

(a) Binary image

OR

- 1(b) Consider the simple 4 x 8, 8-bit image : 4 CO2 D3

20	20	20	95	160	230	230	230
20	20	20	95	160	230	230	230
20	20	20	95	160	230	239	230
20	20	20	95	160	230	230	230

1. Compute the entropy of the image.
 2. Compress the image using Huffman coding.
 3. Compute the compression achieved and the effectiveness of the Huffman coding.
- 1(c) If the original image is 256x256, 8 bits/pixel, it would occupy 65,536 bytes. After compression it occupies 6554/bytes. Calculate the compression ratio. 3 CO2 D3
- 2(a) Describe in details how color based segmentation can be used in an attempt to segment "same-colored" object in real color images. What problems arise? What are some ways of dealing with them? 4 CO2 D3

OR

- 2(a) In the pattern recognition and classification there are four basic steps involved, which are sensing an image, segmenting the image, extracting the features from the segmented 4 CO2 D3

objects, and classification to recognize the specific object. Based on this procedure explain how you can segment and classify a specific object from an 8 bit color image.

- 2(b) The arithmetic decoding process is the reverse of the encoding procedure. Decode the message 0.23355 given the coding model. 4 CO2 D3

Symbol	Probability
a	0.2
e	0.3
i	0.1
O	0.2
u	0.1
!	0.1

- 2(c) Perform the erosion and dilation using the following image and SE. 2 CO2 D3

	1		
1	1	1	
1	1		
	1	1	

Image

1
1
1

SE [centered 1 indicate origin]

OR

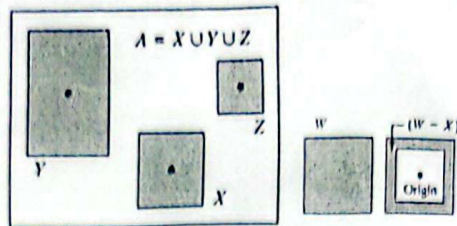
- 2(c) What does rotational invariant means in chain code? How can we make the chain code rotational invariant? 2 CO2 D3 3

Group - B

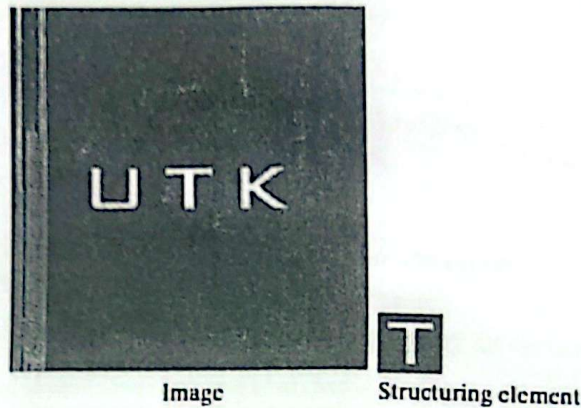
- 3(a) "There are two types of segmentation algorithms- discontinuity and similarity"- Explain both of them with necessary examples. 3 CO3 D3
- 3(b) Write down the formulation of region in region oriented segmentation. 2 CO1 D2
- 3(c) Different types of edges exist in digital images, and various edge detection methods are used to detect them. Write which edge detection method with the type of edge it is most suitable for? Give example for each. 5 CO4 D3 D3

OR

- 3(c) Compare and analyze the *Region Growing* and *Region Splitting & Merging* algorithms used in image segmentation. Explain the steps of each algorithm and illustrate with suitable examples how they work in segmenting an image. 5 CO4 D3 D3
- 4(a) Apply $(A \ominus X) \cap (A^c \ominus (W + X))$ morphological operation on the following image. 5 CO2 D3 D3



- 4(a) Sketch the result of applying the hit-or-miss transform to the image below, using the SE shown. Indicate clearly the origin and border you selected for the structuring element. 5 CO2 D3



- 4(b) Consider the following image and filter. Now extract the features using convolution. Use stride=1. Then reduce the dimensionality of the features vector using max polling. 3 CO1 D2

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Image

1	0	1
0	1	0
1	0	1

Filter

- 4(c) Define signature and polygonal approximation. 2 CO1 C1
- 5(a) In the pattern recognition and classification there are four basic steps involved, which are sensing an image, segmenting the image, extracting the features from the segmented objects, and classification to recognize the specific object. Based on this procedure explain how you can segment and classify a specific object from an 8 bit color image. 5 CO2 C3
- 5(b) What are the main significant of different operators i.e., Sobel in image processing? How these operators are used in the convolutional process to extract the features? How convolutional process plays a significant role in the convolutional neural network? Explain and justify your answer with a 4x4 8 bit color image. 5 CO3 D4